

Woodpeckers and toucans

PHYLUM	Chordata
CLASS	Aves
ORDER	Piciformes
FAMILIES	9
SPECIES	447

This group of arboreal birds includes woodpeckers, toucans, barbets, jacamars, honeyguides, and puffbirds. They all have a similar type of foot, with 2 toes pointing forward and 2 backward, which helps them climb easily. All the birds in this group also

nest in holes. Woodpeckers and barbets carve out their own nest holes, the woodpeckers using their stout bill as a chisel. These are mainly tropical birds; only woodpeckers are also widespread in temperate areas. Curiously, not a single species occurs in Australia. Toucans and barbets help disperse the seeds of some trees by eating their seeds and then passing them through their body, while woodpeckers limit the spread of some tree diseases by feeding on the insects that carry them.

Anatomy

The main climbers in this group, woodpeckers and barbets, have strong leg muscles and use their tail for support when holding themselves vertically upright on tree trunks; woodpeckers have specially stiffened tail feathers for this purpose. Chiseling involves delivering powerful blows to the wood surface, so the skull of woodpeckers is unusually thick to absorb the shock; the nostrils are also slitlike to prevent woodchips from entering the airways. Woodpeckers also have a highly distensible, slightly barbed tongue that is coated with a sticky substance for catching insects. Honeyguides have thick skin to protect them from bee stings and specialized bacteria in their gut that enable them to digest beeswax.



WOODPECKER



BARBET



TOUCAN



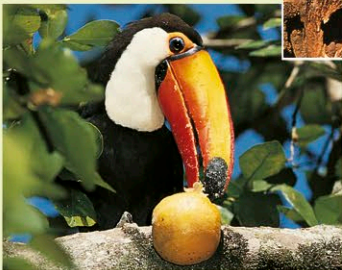
PUFFBIRD

BILL SHAPE
There is great variation in bill shape in this group. A woodpecker's long, sharp bill is ideal for chiseling out wood, while the stout bill of barbets is used to handle animal prey. The large, serrated bill of toucans is adapted for grasping and tearing fruit. Puffbirds and jacamars have a sharp-tipped bill for catching flies.



FEET
The feet of woodpeckers and all their relatives are described as zygodactylous. Two toes point forward and 2 back, an arrangement that helps them climb and perch on tree trunks.

GUIDING
Honeyguides, such as the lesser honeyguide (right), take their name from their habit of leading large mammals to bees' nests.



EATING FRUIT
Toucans, such as the toco toucan (left), eat mainly fruit. The long bill helps them take fruit from otherwise out-of-reach, slender branches.

Reproduction

Although all birds in this group nest in holes, not all use living trees. The smaller woodpeckers prefer to bore into the softer bark of dead and decaying trees. Jacamars, puffbirds, and barbets may dig out burrows, either in the ground or in termite mounds. Unlike the other families, the honeyguides are brood parasites, laying their eggs in the nests of other birds, including woodpeckers. The heels of young toucans are fitted with pads of spikelike projections that protect the ankles from damage against the unlined nest floor.



NESTING
Like most barbets, the black-collared barbet nests in dead trees, sometimes communally. A few species burrow into banks or termite mounds.

NESTING AND FEEDING YOUNG
Woodpeckers use their sturdy bill to bore their own nest holes in tree trunks and defend them vigorously against other hole-nesters. Here, a green woodpecker is feeding its young with ants that it has collected from the ground.